

TURDA, Romania

WMO: 151430

Lat: 46.5831N

Lon: 23.7911E

Elev: 428

StdP: 96.29

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.7	-9.7	-14.8	1.1	-9.6	-12.9	1.3	-8.0	10.7	-4.6	8.6	-2.7	1.0	10	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.9	30.8	19.8	29.1	19.4	27.6	19.0	21.4	28.3	20.7	27.3	19.9	26.1	1.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.6	24.5	18.3	13.9	23.5	17.7	13.3	22.7	64.1	28.4	61.5	27.6	58.7	26.2	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	6.5	5.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-2.7	1.1	5.7	10.9	15.9	19.6	20.2	20.8	16.1	10.6	5.1	-0.5
	DBStd	9.00	4.13	4.45	4.53	4.91	3.51	3.03	2.88	3.17	4.24	4.01	4.25	3.66
	HDD10.0	1367	395	251	144	49	3	0	0	0	2	43	154	327
	HDD18.3	3209	653	483	392	225	89	21	15	13	94	241	397	585
	CDD10.0	1467	0	1	11	74	187	289	318	334	185	61	7	0
	CDD18.3	267	0	0	0	1	15	61	74	89	27	0	0	0
	CDH23.3	2039	0	0	0	22	117	451	528	726	193	2	0	0
	CDH26.7	554	0	0	0	1	13	109	133	253	45	0	0	0
Wind	WSAvg	1.6	1.2	1.7	2.0	2.2	1.9	1.8	1.7	1.4	1.5	1.4	1.3	1.3
Precipitation	PrecAvg	537	22	18	25	44	64	90	85	49	54	43	32	30
	PrecMax	730	44	45	71	75	106	191	136	91	141	89	70	54
	PrecMin	345	7	7	5	4	21	18	16	3	6	1	0	4
	PrecStd	101	11	10	18	19	24	36	33	25	31	24	15	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	13.8	21.1	25.7	28.5	31.0	32.1	33.4	30.5	23.2	16.8	9.7
		MCWB	5.1	8.7	11.2	14.8	16.7	20.8	20.9	20.6	17.8	14.8	12.1	7.6
	2%	DB	4.8	10.9	17.6	23.4	26.3	29.4	29.8	31.3	27.8	21.4	14.3	7.1
		MCWB	3.5	7.6	9.7	13.9	16.8	20.1	20.4	19.3	17.7	14.1	10.7	5.8
	5%	DB	3.5	8.9	14.6	21.1	24.5	27.9	28.0	29.5	25.5	19.4	12.7	5.5
		MCWB	2.4	6.4	8.7	13.0	16.7	19.5	19.6	19.3	16.7	13.4	9.9	4.3
	10%	DB	2.2	7.0	12.1	18.5	22.5	26.1	26.3	27.5	23.2	17.0	11.2	3.9
		MCWB	1.4	4.9	7.1	11.9	15.3	18.9	19.1	19.1	16.2	12.5	8.8	3.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.9	10.0	11.9	16.8	19.2	22.1	22.2	22.0	20.2	16.1	12.7	8.3
		MCDB	7.5	12.9	19.0	22.4	25.3	28.7	28.2	29.8	27.3	20.2	15.4	9.5
	2%	WB	3.8	8.1	10.4	15.2	17.9	21.2	21.4	21.1	18.9	14.9	11.5	6.0
		MCDB	4.7	10.5	16.1	21.5	23.7	27.9	27.7	28.9	24.7	19.7	14.0	7.1
	5%	WB	2.6	6.6	9.0	13.9	17.1	20.3	20.6	20.3	17.8	13.8	10.2	4.5
		MCDB	3.3	8.8	13.9	20.2	22.9	26.6	26.7	27.3	23.4	18.6	12.5	5.4
	10%	WB	1.5	5.2	7.6	12.6	16.2	19.4	19.7	19.5	16.8	12.7	9.2	3.1
		MCDB	2.1	7.1	11.5	17.6	21.2	24.9	25.1	25.9	22.0	16.6	11.3	3.8
Mean Daily Temperature Range	5% DB	MDBR	5.0	6.6	8.8	10.3	9.9	10.3	10.4	10.9	9.7	9.1	6.2	4.2
		MCDBR	5.5	8.9	12.6	13.8	12.8	12.8	12.6	13.9	13.2	13.1	8.2	6.4
		MCWBR	4.7	6.0	6.8	6.5	5.7	5.3	5.1	4.8	4.9	6.5	5.0	5.2
	5% WB	MCDBR	5.2	8.4	11.8	12.5	11.7	12.2	11.5	12.6	11.6	11.9	7.8	6.4
		MCWBR	4.6	5.8	6.5	6.3	5.7	5.5	5.2	5.0	5.0	6.2	4.8	5.2
Clear-Sky Solar Irradiance	taub	0.306	0.332	0.371	0.416	0.402	0.399	0.417	0.422	0.384	0.357	0.333	0.308	
	taud	2.358	2.322	2.271	2.183	2.278	2.311	2.265	2.250	2.329	2.391	2.409	2.386	
	Ebn at Noon	780	831	847	838	864	867	844	823	824	793	742	734	
	Edh at Noon	74	93	114	137	129	125	130	127	107	87	71	65	
All-Sky Solar Radiation	RadAvg	1.22	2.03	3.18	4.35	5.29	5.80	5.70	5.26	3.73	2.55	1.45	0.95	
	RadStd	0.17	0.25	0.44	0.44	0.50	0.48	0.50	0.49	0.42	0.31	0.20	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (69 neighbors)	+0.72	N/A	N/A	+0.94	N/A	N/A	-145	-208	+90	+24

Nomenclature: See separate page